Notes on Gauge Theory

Bob

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1 Principal Bundles and Connections

1.1 Principal Bundles

Principal bundles are both a special type of fiber bundle and the most important one. In the following, we will introduce the definition of principal bundles and some basic structures on them.

Definition 1.1 (G-action)**.** A **smooth right action** of a Lie group G on a manifold P is a smooth map

$$\mu : M \times G \to M,$$

denoted by $\mu(x, g) := x \cdot g$, such that for all $x \in M$ and $g, h \in G$,

$$x \cdot e = x, \quad (x \cdot g) \cdot h = x \cdot (gh).$$

where e is the identity element of G. A manifold M with a smooth right G-action is called a (**right**) G**-manifold**.

We say the action is **free** if for all $x \in M$, $x \cdot g = x$ implies $g = e$.

Definition 1.2 (G-equivariant map)**.** Let M and N be two G-manifolds. A smooth map $f : M \to N$ is called G**-equivariant** if for all $x \in M$ and $g \in G$,

$$f(x \cdot g) = f(x) \cdot g.$$

Definition 1.3 (Principal G-bundle)**.** A **principal** G**-bundle** is a fiber bundle $\pi : P \to M$ with fiber G together with a free right G-action on P such that for each $x \in M$, the local trivialization

$$\phi_U : \pi^{-1}(U) \to U \times G,$$

is G-equivariant, where the right G-action on $U \times G$ is given by

$$(x, h) \cdot g = (x, hg), \quad \forall x \in U, g, h \in G.$$

By a **local trivialization** we mean a open neighborhood U of x and a diffeomorphism $\phi_U : \pi^{-1}(U) \to U \times G$ satisfying the following commutative diagram:

$$\begin{tikzcd}
\pi^{-1}(U) \arrow[rr, "\phi_U"] \arrow[dr, "\pi"'] & & U \times G \arrow[dl, "\mathrm{pr}_1"] \\
& U &
\end{tikzcd}$$

where $\mathrm{pr}_1 : U \times G \to U$ is the projection map.

Example 1.4. Let G be a Lie group and M be a smooth manifold, then the trivial bundle $p_1 : M \times G \to M$ is a principal G-bundle over M.

Example 1.5. If G is a Lie group and M is a smooth manifold, G acts on M freely and properly, then the quotient space M/G is a smooth manifold and the projection map $\pi : M \to M/G$ is a principal G-bundle over M/G. Especially, if G is a finite group equipped with the discrete topology, then a principal G-bundle is exactly a covering space.

Example 1.6. Let G be a Lie group and H be a closed subgroup of G. Then the quotient space G/H is a smooth manifold and the projection map $\pi : G \to G/H$ is a principal H-bundle over G/H. For example, the Hopf fibration $S^1 \to S^3 \to S^2$.

Example 1.7. Let $\pi : E \to M$ be a rank m real (complex) vector bundle over M. Then the frame bundle $\mathrm{Fr}(E)$ is a principal $\mathrm{GL}(m, \mathbb{R})$-bundle ($\mathrm{GL}(m, \mathbb{C})$-bundle) over M.

Example 1.8. Let $\pi : E \to M$ be a rank m real (complex) vector bundle over M. Then the automorphism bundle $\mathrm{Aut}(E)$ is a principal $\mathrm{GL}(m, \mathbb{R})$-bundle ($\mathrm{GL}(m, \mathbb{C})$-bundle) over M. But $\mathrm{Aut}(E)$ is not isomorphic to $\mathrm{Fr}(E)$ in general.

Cocycle Definition of Principal Bundles (Čech Cohomology Perspective)

Proposition 1.9. *Let $\pi : P \to M$ be a principal G-bundle. Then the right G-action on P is transitive on each fiber.*

Proof. Since G acts transitively on $\{x\} \times G$ and the fiber diffeomorphism $\phi_{U,x} : P_x \to G$ is G-equivariant, thus G acts transitively on P_x. □

Lemma 1.10. *For any group G, all right G-equivariant maps are exactly the left translations.*

Proof. Let $f : G \to G$ be a right G-equivariant map, then for all $g \in G$,

$$f(g) = f(e \cdot g) = f(e) \cdot g = l_{f(e)}(g),$$

where $l_{f(e)} : G \to G$ is the left translation by $f(e)$. □

Definition 1.11 (Cocycle definition)**.** A **principal G-bundle** is a fiber bundle $\pi : P \to M$ with fiber G together with a local trivialization $\{(U_\alpha, \phi_\alpha)\}$ such that for each $U_{\alpha\beta} := U_\alpha \cap U_\beta \neq \emptyset$, there exists a smooth map $g_{\alpha\beta} : U_{\alpha\beta} \to G$ such that

$$\phi_\alpha \circ \phi_\beta^{-1}(x, h) = (x, g_{\alpha\beta}(x)h), \quad \forall (x, h) \in U_{\alpha\beta} \times G.$$

In addition, the transition functions $\{g_{\alpha\beta}\}$ satisfy the following cocycle condition:

$$g_{\alpha\beta}(x)g_{\beta\gamma}(x) = g_{\alpha\gamma}(x), \quad \forall x \in U_\alpha \cap U_\beta \cap U_\gamma \neq \emptyset.$$

Remark. By Lemma 1.10, one can easily check that the two definitions of principal bundles are equivalent. These two definitions characterize G-principal bundles from different perspectives.

Example 1.12. Let E be a rank m real (complex) vector bundle over M, with a local trivialization $\{(U_\alpha, \psi_\alpha)\}$ and transition functions $\{g_{\alpha\beta}\}$, then the frame bundle $\mathrm{Fr}(E)$ is a principal $\mathrm{GL}(m, \mathbb{R})$-bundle ($\mathrm{GL}(m, \mathbb{C})$-bundle) over M with local trivialization $\{(U_\alpha, \phi_\alpha)\}$, where $\phi_\alpha : \pi_{\mathrm{Fr}}^{-1}(U_\alpha) \to U_\alpha \times \mathrm{GL}(m, \mathbb{R})$ is defined by

Fundamental Vector Fields

Suppose P is a G-manifold, then for each $A \in \mathfrak{g}$, we can define a vector field $\underline{A}$ on P by

$$\underline{A}_p := \frac{d}{dt}\bigg|_{t=0} p \cdot \exp(tA), \quad \forall p \in P.$$

The vector field $\underline{A}$ is called the **fundamental vector field** associated to A.

Fact 1.1. For each $A \in \mathfrak{g}$, $\underline{A}$ is a C^∞ vector field on P.

The fundamental vector field construction gives rise to a map

$$\sigma : \mathfrak{g} \to \mathfrak{X}(P),^{1} \quad \sigma(A) := \underline{A}.$$

For $p \in P$, define $j_p : G \to P$ by $j_p(g) := p \cdot g$, then $j_{p*} := j_{p*e}$ is

$$j_{p*}(A) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} j_p(\exp(tA)) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} p \cdot \exp(tA) = \underline{A}_p.$$

This gives another discription of the fundamental vector field: $\underline{A}_p = j_{p*}(A)$. It also implies that the map $\sigma : \mathfrak{g} \to \mathfrak{X}(P)$ is $\mathbb{R}$-linear. In fact, we have

Fact 1.2. The map $\sigma : \mathfrak{g} \to \mathfrak{X}(P)$ is a Lie algebra homomorphism, i.e.,

$$[\underline{A}, \underline{B}] = \underline{[A, B]}, \quad \forall A, B \in \mathfrak{g}.$$

Fact 1.3. Let P be a G-manifold, then the fundamental vector fields $\underline{A}$ vanishes at p if and only if $A \in \mathrm{Lie}(\mathrm{Stab}(p))$. Thus for any $p \in P$, the kernel of $j_{p*} : \mathfrak{g} \to T_pP$ is exactly $\mathrm{Lie}(\mathrm{Stab}(p))$. In particular, if the G-action on P is free, then j_{p*} is injective for all $p \in P$.

Example 1.13. Let G acts on itself by right translation, then the fundamental vector field $\underline{A}$ is exactly the left-invariant vector field generated by A.

For $g \in G$, let $c_g : G \to G$ be the conjugation map $c_g(h) := ghg^{-1}$. Then the differential of c_g at the identity element e is the adjoint representation $\mathrm{Ad}_g := (c_g)_* : \mathfrak{g} \to \mathfrak{g}$.

Proposition 1.14. *For each $A \in \mathfrak{g}$ and $g \in G$, we have*

$$(r_g)_* \underline{A} = \underline{\mathrm{Ad}_{g^{-1}}(A)}.$$

where $r_g : P \to P$ is the right translation by g, i.e., $r_g(p) := p \cdot g$.

Proof. For each $p \in P$ and $h \in G$, we have

$$(r_g \circ j_p)(h) = r_g(p \cdot h) = (p \cdot h) \cdot g = pg \cdot (g^{-1}hg) = (j_{pg} \circ c_{g^{-1}})(h).$$

By the chain rule,

$$(r_g)_*(\underline{A}_p) = (r_g)_* j_{p*}(A) = j_{pg*}(c_{g^{-1}})_*(A) = j_{pg*}(\mathrm{Ad}_{g^{-1}}(A)) = \underline{\mathrm{Ad}_{g^{-1}}(A)}_{pg}. \qquad \square$$

[1] $\mathfrak{X}(P) := C^\infty(P, TP)$ is the space of smooth vector fields on P.

1.1.1 Vertical Subbundle of TP

Let $\pi : P \to M$ be a principal G-bundle, at each $p \in P$, the differential $\pi_{*p} : T_pP \to T_{\pi(p)}M$ is surjective. We define $\mathcal{V}_p := \ker \pi_{*p} \subset T_pP$ to be the **vertical tangent space** at p. Then $\mathcal{V} := \bigsqcup_{p\in P} \mathcal{V}_p$ is a subbundle of TP, called the **vertical subbundle** of TP. An element of $\mathcal{V}_p$ is called a **vertical tangent vector** at p.

We have a short exact sequence of vector spaces

$$0 \longrightarrow \mathcal{V}_p \hookrightarrow T_pP \xrightarrow{\pi_{*p}} T_{\pi(p)}M \longrightarrow 0$$

and

$$\dim \mathcal{V}_p = \dim T_pP - \dim T_{\pi(p)}M = \dim G.$$

Proposition 1.15. *For any $A \in \mathfrak{g}$, the fundamental vector field $\underline{A}$ is vertical at every point $p \in P$.*

Proof. Since $(\pi \circ j_p)(g) = \pi(p \cdot g) = \pi(p)$ is a constant map for all $g \in G$, we have $\pi_{*p}(\underline{A}_p) = \pi_{*p}j_{p*}(A) = (\pi \circ j_p)_*(A) = 0$. $\square$

This proposition implies that $j_{p*}(\mathfrak{g}) \subset \mathcal{V}_p$ for all $p \in P$. In fact, we have the following stronger result.

Proposition 1.16. *For each $p \in P$, the map $j_{p*} : \mathfrak{g} \to \mathcal{V}_p$ is an isomorphism.*

Proof. Since G acts freely on P, the map j_{p*} is injective. On the other hand, $\dim \mathcal{V}_p = \dim G = \dim \mathfrak{g}$, thus j_{p*} is an isomorphism. $\square$

Corollary 1.17. *The vertical vectors at $p \in P$ are exactly the fundamental vector at p. Moreover, the vertical subbundle $\mathcal{V}$ of TP is isomorphic to the trivial bundle $P \times \mathfrak{g}$.*

Proof. The first statement follows from the previous proposition. For the second statement, let $A_1, \ldots, A_l$ be a basis of $\mathfrak{g}$, where $l = \dim G$, then $\underline{A}_1, \ldots, \underline{A}_l$ is a global frame of $\mathcal{V}$, thus $\mathcal{V}$ is a trivial bundle. $\square$

1.2 Connections on Principal Bundles

1.2.1 Horizontal Distributions

The map $\pi_* : TP \to TM$ induces a bundle map $\tilde{\pi}_* : TP \to \pi^*TM$ over P. Then we have a short exact sequence of vector bundles over P:

$$0 \longrightarrow \mathcal{V} \hookrightarrow TP \xrightarrow{\tilde{\pi}_*} \pi^*TM \longrightarrow 0$$

In the category of smooth vector bundles, all short exact sequence of vector bundles splits. Thus there exists a smooth subbundle $\mathcal{H}$ of TP such that $TP = \mathcal{V} \oplus \mathcal{H}$.

Definition 1.18. Let $\pi : P \to M$ be a principal G-bundle, a **horizontal distribution** of TP is a smooth subbundle $\mathcal{H}$ of TP such that

- $TP = \mathcal{V} \oplus \mathcal{H}$,
- $\mathcal{H}$ is **right-invariant**, i.e., $r_{g*}\mathcal{H} = \mathcal{H}$, $\forall\, g \in G$, where $r_g : P \to P$, $r_g(p) = p \cdot g$.

Remark. In general, there is no canonical choice of $\mathcal{H}$. A choice of $\mathcal{H}$ is a splitting $k : \pi^*TM \to TP$ such that $\tilde{\pi}_* \circ k = \mathrm{id}_{\pi^*TM}$, together with a G-invariance condition.

$$\begin{tikzcd} 0 \arrow[r] & \mathcal{V} \arrow[r, hook] & TP \arrow[r, shift left, "\tilde{\pi}_*"] & \pi^*TM \arrow[l, shift left, "k"] \arrow[r] & 0 \end{tikzcd}$$

1.2.2 Ehresmann Connections

Let $\pi : P \to M$ be a principal G-bundle, $\mathcal{H}$ be a horizontal distribution of TP. We define the projection map [1]

$$\mathrm{pr}_{\mathcal{V}} : TP = \mathcal{V} \oplus \mathcal{H} \to \mathcal{V},$$

Now we have a splitting:

$$0 \longrightarrow \mathcal{V} \underset{\mathrm{pr}_{\mathcal{V}}}{\overset{\iota}{\rightleftarrows}} TP \xrightarrow{\tilde{\pi}_*} \pi^*TM \longrightarrow 0$$

where $\iota : \mathcal{V} \hookrightarrow TP$ is the inclusion map. We define a $\mathfrak{g}$-valued 1-form ω on P by

$$\omega_p := j_{p*}^{-1} \circ \mathrm{pr}_{\mathcal{V}} : T_pP \xrightarrow{\mathrm{pr}_{\mathcal{V}}} \mathcal{V}_p \xrightarrow{j_{p*}^{-1}} \mathfrak{g}.$$

In terms of ω, the vertical component of $Y_p \in T_pP$ is given by

$$\mathrm{pr}_{\mathcal{V}}(Y_p) = j_{p*}(\omega_p(Y_p)).$$

In summary, given a projection $\mathrm{pr}_{\mathcal{V}} : TP \to \mathcal{V}$, we induce the following composition of maps:

$$\omega : TP \xrightarrow{\mathrm{pr}_{\mathcal{V}}} \mathcal{V} \xrightarrow{\cong} P \times \mathfrak{g} \xrightarrow{\mathrm{pr}_2} \mathfrak{g} \quad \rightsquigarrow \quad \omega \in \Omega^1(P, \mathfrak{g}).$$

So the information of the horizontal distribution $\mathcal{H}$ is encoded in ω.

Theorem 1.19. *If $\mathcal{H}$ is a smooth, right-invariant horizontal distribution, then the $\mathfrak{g}$-valued 1-form ω defined above satisfies the following properties:*

- $\omega_p(\underline{A}_p) = A,\ \forall\, p \in P,\ A \in \mathfrak{g}.$
- *(G-equivariance)* $(r_g)^*\omega = (\mathrm{Ad}_{g^{-1}})\omega,\ \forall\, g \in G.$
- *ω is smooth.*

We say a $\mathfrak{g}$-valued 1-form ω on P satisfying the above properties is an ***Ehresmann connection*** *or simply a* ***connection*** *on P.*

Proof. *1.* Since $\underline{A}$ is already vertical,

$$\omega_p(\underline{A}_p) = j_{p*}^{-1}(\underline{A}_p) = A.$$

2. For each $p \in P$ and $Y_p \in T_pP$, we need to show

$$\omega_{pg}((r_g)_*Y_p) = (\mathrm{Ad}_{g^{-1}})\omega_p(Y_p).$$

Since both side are linear in Y_p, we only need to check the above equality for $Y_p \in \mathcal{V}_p$ and $Y_p \in \mathcal{H}_p$ respectively.

If $Y_p \in \mathcal{V}_p$, then $Y_p = \underline{A}_p$ for some $A \in \mathfrak{g}$, thus

$$\omega_{pg}(r_{g*}\underline{A}_p) = \omega_{pg}(\underline{\mathrm{Ad}_{g^{-1}}(A)}_{pg}) = \mathrm{Ad}_{g^{-1}}(A) = (\mathrm{Ad}_{g^{-1}})\omega_p(\underline{A}_p).$$

If $Y_p \in \mathcal{H}_p$, by the right-invariance of $\mathcal{H}$, we have $r_{g*}(Y_p) \in \mathcal{H}_{pg}$, thus

$$\omega_{pg}(r_{g*}Y_p) = 0 = (\mathrm{Ad}_{g^{-1}})\omega_p(Y_p).$$

3. Omit. ☐

[1] Although $\mathcal{V}$ is intrinsically defined, the vertical component of a tangent vector depends on the choice of $\mathcal{H}$.

Viewing ω as a map from TP to $\mathfrak{g}$. Both TP and $\mathfrak{g}$ are G-manifolds, where

$$g \cdot Y_p := (r_g)_* Y_p, \quad g \cdot A := \mathrm{Ad}_{g^{-1}}(A).$$

Then the G-equivariance condition of ω

$$\omega_{pg}((r_g)_* Y_p) = (\mathrm{Ad}_{g^{-1}})\omega_p(Y_p),$$

can be rephrased as $\omega : TP \to \mathfrak{g}$ is a G-equivariant map.

Example 1.20. The Maurer-Cartan form ω_{MC} on a Lie group G is the unique $\mathfrak{g}$-valued 1-form on G such that

- $\omega_{\mathrm{MC}}(A) = A$ for all $A \in \mathfrak{g} = T_e G$,
- ω_{MC} is left-invariant, i.e., $l_g^* \omega_{\mathrm{MC}} = \omega_{\mathrm{MC}}$ for all $g \in G$.

Let ω_{MC} be the Maurer-Cartan form on G, then $\pi^* \omega_{\mathrm{MC}}$ is an Ehresmann connection on the trivial bundle $\pi : M \times G \to M$.

Remark. Let ω be an Ehresmann connection on P, for any $p \in P$, $j_p : G \to P$ pulls back ω to ω_{MC} on G, i.e., $j_p^* \omega = \omega_{\mathrm{MC}}$.

The Horizontal Distribution of an Ehresmann Connection

In the previous section we have shown that a smooth, right-invariant horizontal distribution $\mathcal{H}$ on P induces an Ehresmann connection ω. We now prove the converse.

Theorem 1.21. *If ω is an Ehresmann connection on the principal G-bundle $\pi : P \to M$, then $\mathcal{H}_p := \ker \omega_p$, $p \in P$ defines a smooth, right-invariant horizontal distribution on P.*

Proof. We need to verify three properties:
1. For each $p \in P$, there is an exact sequence

$$0 \longrightarrow \mathcal{H}_p \hookrightarrow T_p P \xrightarrow{\omega_p} \mathfrak{g} \longrightarrow 0$$

The map $j_{p*} : \mathfrak{g} \to \mathcal{V}_p \subset T_p P$ provides a splitting of the above sequence, thus

$$T_p P \cong \mathfrak{g} \oplus \mathcal{H}_p \cong \mathcal{V}_p \oplus \mathcal{H}_p.$$

2. Suppose $Y_p \in \mathcal{H}_p$, then $\omega_p(Y_p) = 0$. By the G-equivalence of ω, we have

$$\omega_{pg}(r_{g*} Y_p) = (r_g^* \omega)_p(Y_p) = (\mathrm{Ad}_{g^{-1}})\omega_p(Y_p) = 0.$$

Hence $r_{g*} Y_p \in \mathcal{H}_{pg}$.
3. $\mathcal{H}$ is a smooth distribution. We omit the details here. □

1.2.3 Horizontal Lift and Parallel Transport

Let $\pi : P \to M$ be a principal G-bundle and ω be an Ehresmann connection on P. For each (piecewise) smooth curve $\gamma : [0,1] \to M$ and $p \in P$ such that $\pi(p) = \gamma(0)$, there exists a unique (piecewise) smooth curve $\tilde{\gamma} : [0,1] \to P$ such that

$$\tilde{\gamma}(0) = p, \quad \pi \circ \tilde{\gamma} = \gamma, \quad \text{and} \quad \tilde{\gamma}'(t) \in \mathcal{H}_{\tilde{\gamma}(t)}, \forall t \in [0,1].$$

Definition 1.22. We call $\tilde{\gamma}$ the **horizontal lift** of γ starting at p. If we denote $\tilde{\gamma}_p$ to be the horizontal lift of γ starting at p, then the map

$$P^\gamma : \mathrm{Fiber}_{\gamma(0)} \to \mathrm{Fiber}_{\gamma(1)}, \quad P^\gamma(p) := \tilde{\gamma}_p(1),$$

is called the **parallel transport** along γ.

Proposition 1.23. *Let $\tilde{\gamma}_p$ be the horizontal lift of γ starting at p, then $\tilde{\gamma}_{pg} = \tilde{\gamma}_p \cdot g$ for any $g \in G$.*

Proof. By the right-invariance of $\mathcal{H}$, we have

$$(\tilde{\gamma}_p \cdot g)(0) = pg, \quad \pi \circ (\tilde{\gamma}_p \cdot g) = \pi \circ \tilde{\gamma}_p = \gamma \quad \text{and} \quad (\tilde{\gamma}_p \cdot g)'(t) = r_{g*}\tilde{\gamma}_p'(t) \in \mathcal{H}_{(\tilde{\gamma}_p \cdot g)(t)}, \forall t \in [0,1].$$

Thus $\tilde{\gamma}_p \cdot g$ is the horizontal lift of γ starting at pg, i.e., $\tilde{\gamma}_{pg} = \tilde{\gamma}_p \cdot g$. $\square$

Corollary 1.24. *The parallel transport P^γ is a diffeomorphism between the fibers $\mathrm{Fiber}_{\gamma(0)}$ and $\mathrm{Fiber}_{\gamma(1)}$.*

For any two points $x, y \in M$, we can choose a curve γ connecting x and y, then the parallel transport P^γ gives a diffeomorphism between the fibers Fiber_x and Fiber_y. In this sense, a connection provides a way to identify fibers over different points of the base manifold.

Like the case of affine connections on vector bundles, a family of parallel transports along all curves in M also determines a unique Ehresmann connection on P.

1.2.4 Connections on the frame bundle $\mathrm{Fr}(E)$

In the previous sections, we have defined affine connections on vector bundles E. Since the frame bundle $\mathrm{Fr}(E)$ is a natural principal bundle corresponding to E, we can study the relation between connections on E and connections on $\mathrm{Fr}(E)$.

Let ∇ be an affine connection on E. Then for any curve $\gamma : [0,1] \to M$, we have a parallel transport $P_t^\gamma : E_{\gamma(0)} \to E_{\gamma(t)}$ defined by ∇. Since P_t^γ is a linear isomorphism, it induces a diffeomorphism $\tilde{P}_t^\gamma : \mathrm{Fr}(E_{\gamma(0)}) \to \mathrm{Fr}(E_{\gamma(t)})$ by

$$\tilde{P}_t^\gamma(e_1, \ldots, e_m) := (P_t^\gamma(e_1), \ldots, P_t^\gamma(e_m)),$$

where $(e_1, \ldots, e_m) \in \mathrm{Fr}(E_{\gamma(0)})$ is a frame of $E_{\gamma(0)}$. Then $\tilde{\gamma}_e(t) = \tilde{P}_t^\gamma(e_1, \ldots, e_m)$ is a lift of γ starting at $e = (e_1, \ldots, e_m)$.

Definition 1.25. The **horizontal distribution** of $\mathrm{Fr}(E)$ induced by ∇ is defined by

$$\mathcal{H}_e := \{\tilde{\gamma}_e'(0) \mid \gamma : [0,1] \to M, \, \gamma(0) = \pi(e)\},$$

and $\mathcal{H} := \bigsqcup_{e \in \mathrm{Fr}(E)} \mathcal{H}_e$

Proposition 1.26. *$\mathcal{H}$ is a smooth, right-invariant horizontal distribution on* $\mathrm{Fr}(E)$.

Proof. 1. Smoothness: omit. 2. $T_e\mathrm{Fr}(E) = \mathcal{V}_e \oplus \mathcal{H}_e$: omit.

3. Right-invariance: Let $e = (e_1, \ldots, e_m) \in \mathrm{Fr}(E)$ and $a \in \mathrm{GL}(m, \mathbb{R})$, since P_t^γ is a linear isomorphism, we have

$$\tilde{\gamma}_{e\cdot a}(t) = \tilde{P}_t^\gamma\Big((e_1, \ldots, e_m)a\Big) = \big(P_t^\gamma(e_1), \ldots, P_t^\gamma(e_m)\big)a = \tilde{P}_t^\gamma(e_1, \ldots, e_m) \cdot a = \tilde{\gamma}_e(t) \cdot a,$$

Thus $r_{a*}\tilde{\gamma}_e'(0) = \tilde{\gamma}_{e\cdot a}'(0)$, which implies $r_{a*}\mathcal{H}_e = \mathcal{H}_{e\cdot a}$. □

Since now we got a horizontal distribution $\mathcal{H}$ on $\mathrm{Fr}(E)$, we can define an Ehresmann connection on $\mathrm{Fr}(E)$ —— the $\mathfrak{gl}(n, \mathbb{R})$-valued 1-form ω. Recall that given a local trivialization $\{(U_\alpha, \psi_\alpha)\}$ of E, we have a local connection 1-form ω_α on U_α defined by ∇. There is no doubt that ω_α is related to ω in some way.

Lemma 1.27. *Let $\pi : E \to M$ be a rank m bundle with a connetion ∇, $x \in M$ and $e_x \in \mathrm{Fr}(E_x)$. For any smooth curve $\gamma : [0,1] \to M$ starting at x, there is a unique horizontal lift $\tilde{\gamma} :\ [0,1] \to \mathrm{Fr}(E)$ with the initial point $e_x = (e_1, \ldots, e_m)$. Let $s : U \to \mathrm{Fr}(E)$ be a local frame of E around x, with $s(x) = e_x$, then $s(\gamma(t))$ is also a lift of γ starting at e_x. Both $\tilde{\gamma}(t)$ and $s(\gamma(t))$ are frames of $E_{\gamma(t)}$, thus there exists a unique $a(t) \in \mathrm{GL}(m, \mathbb{R})$ such that*

$$s(\gamma(t)) = \tilde{\gamma}(t) \cdot a(t).$$

Then we have the following fomula

$$s_*(\gamma'(0)) = \tilde{\gamma}'(0) + \underline{a'(0)}_{e_x}.$$

Proof. Let $\mu : \mathrm{Fr}(E) \times \mathrm{GL}(m, \mathbb{R}) \to \mathrm{Fr}(E)$ be the right action of $\mathrm{GL}(m, \mathbb{R})$ on $\mathrm{Fr}(E)$. Then

$$s(\gamma(t)) = \mu(\tilde{\gamma}(t), a(t)),$$

with $\gamma(0) = x$, $\tilde{\gamma}(0) = e_x$ and $a(0) = I$.

$$\begin{aligned} s_*(\gamma'(0)) =& \mu_{*(\tilde{\gamma}(0), a(0))}(\tilde{\gamma}'(0), a'(0)) \\ =& \mu_{*(e_x, I)}(\tilde{\gamma}'(0), 0) + \mu_{*(e_x, I)}(0, a'(0)) \\ =& \tilde{\gamma}'(0) + \underline{a'(0)}_{e_x}. \end{aligned}$$

□

Lemma 1.28. *Since $s = (s_1, \ldots, s_m)$ is a local frame of E, we can define a local connection 1-form ω_s of ∇ w.r.t s. Then we have $\omega_s(\gamma'(0)) = a'(0)$.*

Proof. Let $\tilde{\gamma}(t) = (\tilde{\gamma}_1(t), \ldots, \tilde{\gamma}_m(t))$, then $\tilde{\gamma}_i(t)$ is the parallel transport of e_i along γ. Now we have

$$s(\gamma(t))_i = \tilde{\gamma}_j(t) a_i^j(t),$$

take the covariant derivative of both side at $t = 0$, we get

$$\omega_i^j(\gamma'(0)) s_j(\gamma(0)) = \nabla_{\gamma'(0)} s_i = \nabla_{\gamma'(0)}(\tilde{\gamma}_j(t) a_i^j(t)) = a_i^{j\prime}(0)\tilde{\gamma}_j(0).$$

Both $s(\gamma(0))$ and $\tilde{\gamma}(0)$ are $e_x = (e_1, \ldots, e_m)$, thus we have $\omega_s(\gamma'(0)) = a'(0)$. □

Corollary 1.29. *In the notation above, we have*

$$\tilde{\gamma}'(0) = s_*(\gamma'(0)) - \underline{\omega_s(\gamma'(0))}_{e_x}.$$

Corollary 1.30. *Let ∇ be an affine connection on $\pi : E \to M$. Suppose $\{(U_\alpha, s_\alpha)\}$ is a local trivialization of E, then we have a family of connection 1-forms ω_α of ∇ w.r.t s_α. For $x \in U_\alpha$, $p = s_\alpha(x) \in \mathrm{Fr}(E_x)$ and $X_x \in T_xM$, let $\tilde{X}_p \in T_p\mathrm{Fr}(E)$ be the horizontal lift of X_x in $\mathrm{Fr}(E)$, then we have*

$$\tilde{X}_p = (s_\alpha)_*(X_x) - \underline{\omega_\alpha(X_x)}_p.$$

Theorem 1.31. *In the notation above, let ω be the Ehresmann connection on $\mathrm{Fr}(E)$ induced by ∇, then we have*

$$\omega_\alpha = s_\alpha^*\omega.$$

Note that $s_\alpha : U_\alpha \to \mathrm{Fr}(E)$ can be viewed as a local section of $\mathrm{Fr}(E)$.

Proof. Let $x \in U_\alpha$, $p = s_\alpha(x) \in \mathrm{Fr}(E_x)$ and $X_x \in T_xM$. Let $\tilde{X}_p \in T_p\mathrm{Fr}(E)$ be the horizontal lift of X_x in $\mathrm{Fr}(E)$, then we have

$$\tilde{X}_p = (s_\alpha)_*(X_x) - \underline{\omega_\alpha(X_x)}_p.$$

Applying ω_p to both side, we get

$$0 = \omega_p(\tilde{X}_p) = (s_\alpha^*\omega)_x(X_x) - \omega_p(\underline{\omega_\alpha(X_x)}_p) = (s_\alpha^*\omega)_x(X_x) - \omega_\alpha(X_x).$$

Since X_x is arbitrary, we have $\omega_\alpha = s_\alpha^*\omega$. $\square$